

① Show that if $x(t) \Leftrightarrow X(\omega)$

then $x(t-t_0) \Leftrightarrow e^{j\omega t_0} X(\omega)$

Soln: $\mathcal{F}\{x(t-t_0)\} = \int_{-\infty}^{\infty} x(t-t_0) e^{-j\omega t} dt$

→ by change of variable $\tau = t - t_0$, we get

$$\begin{aligned} \mathcal{F}\{x(t-t_0)\} &= \int_{-\infty}^{\infty} x(\tau) e^{-j\omega(\tau+t_0)} d\tau \\ &= e^{-j\omega t_0} \underbrace{\int_{-\infty}^{\infty} x(\tau) e^{-j\omega \tau} d\tau}_{X(\omega)} = e^{-j\omega t_0} X(\omega) \end{aligned}$$

② Show that if $x(t) \Leftrightarrow X(\omega)$

then $x(t) e^{j\omega_0 t} \Leftrightarrow X(\omega - \omega_0)$

Soln

$$\begin{aligned} \mathcal{F}\{x(t) e^{j\omega_0 t}\} &= \int_{-\infty}^{\infty} x(t) e^{j\omega_0 t} e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} x(t) e^{-j(\omega - \omega_0)t} dt = X(\omega - \omega_0) \end{aligned}$$

(2)

③ Verify the duality property i.e

if $x(t) \Leftrightarrow X(\omega)$

then $X(t) \Leftrightarrow 2\pi x(-\omega)$

Soln By Defn. of IFT $x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$

$$\Rightarrow \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega = 2\pi x(t)$$

→ replacing t with $-t$, we get

$$\int_{-\infty}^{\infty} X(\omega) e^{-j\omega t} d\omega = 2\pi x(-t)$$

→ now interchanging t and ω , we get

$$\int_{-\infty}^{\infty} X(t) e^{-j\omega t} dt = 2\pi x(-\omega)$$

$$\underbrace{\int_{-\infty}^{\infty} X(t) e^{-j\omega t} dt}_{\mathcal{F}\{X(t)\}} = 2\pi x(-\omega)$$

$$\Rightarrow X(t) \Leftrightarrow 2\pi x(-\omega)$$

④ (using properties)

Find FT of $x(t) = \frac{1}{a^2 + t^2}$

→ we notice from FT Table that

$$x(t) e^{-a|t|} \Leftrightarrow \frac{2a}{a^2 + \omega^2} X(\omega)$$

→ By Duality (i.e. $X(t) \Leftrightarrow 2\pi x(-\omega)$)

i.e. replace ω with t on RHS, and replace t with $-\omega$ on LHS and multiply by

$$\Rightarrow X(t) = \frac{2a}{a^2 + t^2} \quad \left(\text{since } X(\omega) = \frac{2a}{a^2 + \omega^2} \right)$$

$$\text{and } \Rightarrow x(-\omega) = e^{-a|\omega|} \quad \left(\text{since } x(t) = e^{-a|t|} \right)$$

$$\text{Duality gives: } \frac{2a}{a^2 + t^2} \Leftrightarrow 2\pi e^{-a|\omega|}$$

$$\text{By linearity: } \frac{1}{a^2 + t^2} \Leftrightarrow \frac{2\pi}{2a} e^{-a|\omega|}$$

(4)

⑤ Verify that if $x(t) \Leftrightarrow X(\omega)$
 then $\frac{dx(t)}{dt} \Leftrightarrow j\omega X(\omega)$

Soln using IFT formula

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \quad \text{--- } \textcircled{\star}$$

$$\Rightarrow \frac{dx(t)}{dt} = \frac{1}{2\pi} \frac{d}{dt} \left[\int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \right]$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) \frac{\partial}{\partial t} (e^{j\omega t}) d\omega$$

$$\left[\frac{dx(t)}{dt} \right] = \frac{1}{2\pi} \int_{-\infty}^{\infty} j\omega X(\omega) e^{j\omega t} d\omega \quad \text{--- } \textcircled{\Delta}$$

Comparing $\textcircled{\star}$ & $\textcircled{\Delta}$ tells us that

$$\frac{dx(t)}{dt} \Leftrightarrow j\omega X(\omega)$$

⑥ Find FFT of $X(\omega) = \frac{1}{(a+j\omega)^2}$
using Time convolution theorem.

⑤

Soln Time convolution theorem says

$$\text{if } y(t) = x(t) * h(t)$$

$$\text{then } Y(\omega) = X(\omega) H(\omega)$$

now we have $X(\omega) = \frac{1}{(a+j\omega)^2} = \frac{1}{(a+j\omega)} \frac{1}{(a+j\omega)}$

$\downarrow$
 $X_1(\omega)$

$\downarrow$
 $X_2(\omega)$

$$\Rightarrow X(\omega) = X_1(\omega) X_2(\omega)$$

$$\Rightarrow x(t) = x_1(t) * x_2(t)$$

where $x_1(t) = x_2(t) = e^{-at} u(t)$

since $\left(e^{-at} u(t) \Leftrightarrow \frac{1}{a+j\omega} \right)$

$$\begin{aligned} \Rightarrow x(t) &= e^{-at} u(t) * e^{-at} u(t) \\ &= \int_{-\infty}^{\infty} e^{-az} e^{-a(t-z)} u(z) u(t-z) dz \\ &= e^{-at} \int_0^t dz = t e^{-at} u(t) \end{aligned}$$

recall $\left(f(t) * g(t) = \int_{-\infty}^{\infty} f(z) g(t-z) dz \right)$

(7) Assuming zero ILS, find impulse response of

(6)

$$\frac{dy(t)}{dt} + 2y(t) = x(t) + \frac{dx(t)}{dt}$$

Solve
Taking FT

$$\Rightarrow j\omega Y(\omega) + 2Y(\omega) = X(\omega) + j\omega X(\omega)$$

$$\Rightarrow (j\omega + 2) Y(\omega) = (1 + j\omega) X(\omega)$$

$$Y(\omega) = \underbrace{\frac{(1 + j\omega)}{(2 + j\omega)}}_{H(\omega)} X(\omega)$$

$$\Rightarrow H(\omega) = \frac{1 + j\omega}{2 + j\omega} = \frac{2 + j\omega - 1}{2 + j\omega} = 1 - \frac{1}{2 + j\omega}$$

$\Rightarrow$ Taking IFT to get

$$h(t) = \delta(t) - e^{-2t} u(t)$$

② Prove Parseval's Theorem

⑦

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |x(\omega)|^2 d\omega$$

→ Reading assignment: look for its proof & usage!

⑨ Given system with zero-state

$$\frac{dy(t)}{dt} + 2y(t) = x(t)$$

find $y(t)$ if $x(t) = e^{\pm t} u(t)$

Soln.

$$j\omega Y(\omega) + 2Y(\omega) = X(\omega)$$

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{1}{2 + j\omega}$$

$$\text{also } X(\omega) = \frac{1}{1 + j\omega}$$

$$\Rightarrow Y(\omega) = H(\omega) X(\omega) = \frac{1}{1 + j\omega} \cdot \frac{1}{2 + j\omega}$$

with PFE: $y(t) = (e^{-t} - e^{-2t})u(t)$